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27 July 1967
SUPERSEDING
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11 May 1959

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M31 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M31 Assembly, loaded with Cartridge, Initiator, Delay, M93, for use with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specifications for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	- Ammunition Data Card
MS 28741	- Hose Assembly, Detachable End Fitting, Medium Pressure, Size No. 4

DRAWINGS

Munitions Command

F8595251	- Initiator, Cartridge Actuated, Delay, M31 Assembly
F8798142	- Box, Packing, Ammunition for Initiator, Delay, M10 in Fiber Container M398 (See 5.1)

FSC 1377

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2.1 (Cont'd)

DRAWINGS (Cont'd)

Munitions Command

F8798143 - Container, Fiber, Ammunition M398 for Initiator Delay, M10 (See 5.1)

IEL-8595251 - Inspection Equipment List for Initiator, Delay, M31 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following.

3.2 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual magnetism.- The initiator shall not deflect the indicator of an approved compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size No. 4, in accordance with MS 28741.

TABLE I

Temperature	Pressure (psi)	Ignition Delay (See 6.2)
-65° to -70°F	500 - 1250	1.3 seconds, max.
65° to 75°F	600 - 1500	1.0 ± 0.1 seconds
160° to 165°F	700 - 1700	0.8 seconds, min.

3.4.1.1 Initiation.- The initiator shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, in accordance with MS 28741, at 70° ± 5°F.

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3.4.2 The initiator shall not fail to fire.

3.4.3 The initiator shall function without mechanical failure (See 6.1).

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection Provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

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4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, Delay, M93, Drawing D8593312 covering a detail of Drawing F8595251

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O"-ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O"-ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual

4.2.2.2 Assembly, Drawing F8595251, less Drawings A8359536, B8593298, D8594856, D8594857, D8594860 and D8593312

CRITICAL	
1. Set screw missing	Visual

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4.2.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Drawing F8595251, less Drawings A8359536, B8593298 and D8594857

CRITICAL	
1. Cartridge loose	Manual and Visual
2. Cartridge retainer loose	Manual and Visual
3. Metering groove of cartridge retainer missing	Visual
4. Elongated hole of cartridge retainer missing	Visual
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.4 Assembly, Drawing F10520220, less Drawings A8359536 and B8593298

CRITICAL	
1. Location of cartridge retainer	Gage
2. Shear pin not engaged in firing pin	Gage
3. Any part missing	Visual
4. Improper assembly	Visual
5. Foreign matter in gas port	Visual
MAJOR	
101. Length from firing pin housing boss face to gas port face, max.	Gage

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Varnish missing or inadequate	Visual
202. Location of label incorrect	Visual
203. Barrier material missing from set screw	Visual

4.2.2.5 Assembly, Drawing F8595251

CRITICAL	
1. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Height of cap, max.	Gage
102. Cap set screw missing, loose or above flush	Visual
MINOR	
201. Total length, max.	Gage
202. Location of marking, incorrect	Visual
203. Shipping cap or plug missing	Visual
204. Protective finish inadequate or defective	Visual

4.2.2.6 Container, Fiber, Drawing C8798143

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

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4.2.2.7 Box, Packing, Drawing C8798142

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- The initiator cap and chamber of each assembled initiator shall be subjected to the breakaway torque application. Failure of any assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.3 shall be cause for rejection of the initiator.

4.2.3.3 Functional.- The following test shall be performed by the Government.

4.2.3.3.1 Ballistics.- Twenty-one units shall be tested to determine compliance with the requirements of 3.4.1, 3.4.2 and 3.4.3. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. The variables sampling plan of MIL-STD-414, Example C-2 or C-3, as applicable, for the AQL listed in Table III shall be used for each sample of 7 units. Failure of any unit to comply with the requirements of 3.4.1, 3.4.2 and 3.4.3 or failure of a sample to meet the criteria of the variable sampling plan shall be cause for rejection of the lot.

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4.2.3.3.1 (Cont'd)

TABLE III

Temperature	Sample Size	AQL
-65° to -70°F	7	0.65
65° to 75°F	7	0.65
160° to 165°F	7	0.65

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8595251.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.2 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.2 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.3 Functional.

4.3.3.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.3.3.1.1.

4.3.3.1.1 Temperature conditioning.- A standard M31 Initiator, loaded with a dummy M93 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

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5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8798142 and C8798143, except that marking requirements shall reference M31 Initiator.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous rise on the pressure time record.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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